05 | Residual input edges: one downstream cotangent

First explain r1 itself, then the downstream scalar F = r1^2 / 2.

h1 = 1.0000 + o1 = 2.7616 = r1 = 3.7616

Endpoint h1

0.500	0.500
0.000	0.000

Σ = 1.0000

+

Endpoint o1

0.165	0.165
1.636	1.636

Σ = 3.6015

=

Endpoint r1

0.665	0.665
1.636	1.636

Σ = 4.6015

Identity edge

1.881	1.881
0.000	0.000

Σ = 3.7616

+

Attention edge

0.619	0.619
6.155	6.155

Σ = 13.5475

=

Total for F

2.500	2.500
6.155	6.155

Σ = 17.3091

$$g_r = \nabla_r F; \quad E_{id} = x \odot J_h^T g_r, \quad E_{attn} = x \odot J_o^T g_r$$

Every input-to-F path crosses this add. Using the total gradient at h as identity would count the attention path twice.